

Davide Ottaviani

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Education

Sapienza University of Rome, MS in Artificial Intelligence and Robotics Oct 2025 – Present

- **Coursework:** Advanced Artificial Intelligence, Machine Learning, Robotics, Computer Vision, Natural Language Processing, Reinforcement Learning, Computational Theory, Data Analysis and Modeling

Sapienza University of Rome, BS in Computer and Control Engineering Sept 2022 – Oct 2025

- Final Grade: 110/110
- **Coursework:** Computer Architecture, Algorithms and Complexity, Computational Theory, Fundamentals of Artificial Intelligence, Data Structures and Programming

Projects

Joint Detection of AI-Generated Images & Post-Processing github.com/cv_project_2526

- Computer Vision course project: a dual-task pipeline detecting AI-generated images and identifying their post-processing (original / transmitted / re-digitized), built on a frozen DINOv3 backbone with an FFT frequency branch and LoRA adapters.
- Compared Baseline, Joint, and LoRA training regimes across 6 configurations; best model reached 95.7% real/fake accuracy and 96.4% transformation-type accuracy, with full ablations on loss weighting and shared-trunk bottleneck size.
- Tools Used: PyTorch, DINOv3 (ViT-B/16), LoRA/PEFT, Transformers

Multilingual Semantic Search & Retrieval

- Natural Language Processing course project: an information retrieval system built on a bi-encoder architecture, comparing fine-tuning strategies (Sentence Pairs vs. Triplets) with a cross-encoder reranking stage.
- Fine-tuned DistilBERT and all-MiniLM with MultipleNegativesRankingLoss; best pipeline (Triplets + reranking) reached MRR 0.72, up from a 0.37 baseline.
- Tools Used: Python, SentenceTransformers, PyTorch

Interactive CC8 – Procedural Mars Rover 3D Game

live demo

- Browser-based 3D driving game built from scratch with Three.js: a rover crosses a procedurally generated Martian desert with a physically animated rocker-bogie suspension, a day/night cycle, and a dependency-driven mission system.
- Implemented infinite procedural terrain (simplex-noise fBm with seamless chunk streaming) and live-synthesized audio via the Web Audio API.
- Tools Used: Three.js, JavaScript, WebGL, Vite

Technologies

Languages: Python, C++, C, Java, SQL, JavaScript, MATLAB

AI / ML / NLP / Vision: PyTorch, Transformers, SentenceTransformers, OpenCV, LoRA / PEFT

Robotics: ROS2, Kinematics, 3D Simulation, Three.js

Tools & Other: Git, GitHub, Docker, Linux / Bash, REST APIs, Object-Oriented Programming, Data Structures